

Classification Models Report for Social Media Advertisement

Confusion Matrix

Predicted class ↓	Actual Class →	Non-Purchased	Purchased
Non-Purchased		73	6
Purchased		5	36

Evaluation Metrics for Confusion Matrix

$$\begin{aligned}\text{Accuracy} &= \frac{T(\text{Non-Purchased})+T(\text{Purchased})}{T(\text{Non-Purchased})+T(\text{Purchased})+F(\text{Non-Purchased})+F(\text{Purchased})} \\ &= \frac{73+36}{73+36+6+5} \\ &= \frac{109}{120} \\ &= 0.91\end{aligned}$$

$$\begin{aligned}\text{Recall(N-P)} &= \frac{T(\text{Non-Purchased})}{T(\text{Non-Purchased})+F(\text{Purchased})} \\ &= \frac{73}{73+5} \\ &= \frac{73}{78} \\ &= 0.94\end{aligned}$$

$$\begin{aligned}\text{Recall(P)} &= \frac{T(\text{Purchased})}{T(\text{Purchased})+F(\text{Non-Purchased})} \\ &= \frac{36}{36+6} \\ &= \frac{36}{42} \\ &= 0.86\end{aligned}$$

$$\begin{aligned}
 \text{Precision(N-P)} &= \frac{T(\text{Non-Purchased})}{T(\text{Non-Purchased})+F(\text{Non-Purchased})} \\
 &= \frac{73}{73+6} \\
 &= \frac{73}{79} \\
 &= 0.92
 \end{aligned}$$

$$\begin{aligned}
 \text{Precision(P)} &= \frac{T(\text{Purchased})}{T(\text{Purchased})+F(\text{Purchased})} \\
 &= \frac{36}{36+5} \\
 &= \frac{36}{41} \\
 &= 0.88
 \end{aligned}$$

$$\begin{aligned}
 \text{F1 score(N-P)} &= \frac{2 * \text{Recall(N-P)} * \text{Precision(N-P)}}{\text{Recall(N-P)} + \text{Precision(N-P)}} \\
 &= \frac{2 * 0.94 * 0.92}{0.94 + 0.92} \\
 &= \frac{1.73}{1.86} \\
 &= 0.93
 \end{aligned}$$

$$\begin{aligned}
 \text{F1 score(P)} &= \frac{2 * \text{Recall(P)} * \text{Precision(P)}}{\text{Recall(P)} + \text{Precision(P)}} \\
 &= \frac{2 * 0.86 * 0.88}{0.86 + 0.88} \\
 &= \frac{1.51}{1.74} \\
 &= 0.88
 \end{aligned}$$

$$\begin{aligned}
 \text{Macro Avg.(Pr)} &= \frac{\text{Precision(N-P)} + \text{Precision(P)}}{2} \\
 &= \frac{0.92 + 0.88}{2}
 \end{aligned}$$

$$= \frac{1.8}{2}$$

$$= 0.90$$

$$\text{Macro Avg. (Rc)} = \frac{\text{Recall(N-P)} + \text{Recall(P)}}{2}$$

$$= \frac{0.94 + 0.86}{2}$$

$$= \frac{1.8}{2}$$

$$= 0.90$$

$$\text{Macro Avg. (F1)} = \frac{\text{F1score(N-P)} + \text{F1score(P)}}{2}$$

$$= \frac{0.93 + 0.88}{2}$$

$$= \frac{1.81}{2}$$

$$= 0.91$$

$$\text{Wt Avg. (Pr)} = (\text{Precision(N-P)} * (70/120)) + (\text{Precision(P)} * (41/120))$$

$$= (0.92 * (70/120)) + (0.88 * (41/120))$$

$$= (0.92 * 0.58) + (0.88 * 0.34)$$

$$= 0.53 + 0.29$$

$$= 0.82$$

$$\text{Wt Avg. (Rc)} = (\text{Recall(N-P)} * (70/120)) + (\text{Recall(P)} * (41/120))$$

$$= (0.94 * (70/120)) + (0.86 * (41/120))$$

$$= (0.94 * 0.58) + (0.86 * 0.34)$$

$$= 0.55 + 0.29$$

$$\begin{aligned}
&= 0.84 \\
\text{Wt Avg.}(F1) &= (F1\text{score}(N-P)*(70/120)) + (F1\text{score}(P)*(41/120)) \\
&= (0.93*(70/120)) + (0.88*(41/120)) \\
&= (0.93*0.58) + (0.88*0.34) \\
&= 0.54 + 0.29 \\
&= 0.83
\end{aligned}$$

Random forest Classification Report

Class	Precision	Recall	F1-score	Support
0 (Non-Purchased)	0.94	0.92	0.93	79
1 (Purchased)	0.86	0.88	0.87	41
Accuracy			0.91	120
Macro Average	0.90	0.90	0.90	120
Weighted Average	0.91	0.91	0.91	120

- What is the overall performance of the model Random Forest?
0.91
- What is the correct classification of non-purchased?
0.92
- What is the correct classification of purchased?
0.88
- What is the correct and wrong classification of non-purchased?
0.94
- What is the correct and wrong classification of purchased?
0.86
- What is the overall performance of non-purchased?
0.93
- What is the overall performance of purchased?

0.87

8. What is the average performance of correct and wrong classification?

0.90

9. What is the average performance of correct classification?

0.90

10. What is the average overall performance in classification?

0.90

11. What is the weight of correct and wrong classification?

0.91

12. What is the weight of correct classification?

0.91

13. What is the weight of overall performance in classification?

0.91

Decision Tree Classification Report

Class	Precision	Recall	F1-score	Support
0 (Non-Purchased)	0.96	0.90	0.93	79
1 (Purchased)	0.83	0.93	0.87	41
Accuracy			0.91	120
Macro Average	0.89	0.91	0.90	120
Weighted Average	0.91	0.91	0.91	120

1. What is the overall performance of the model Decision Tree?

0.91

2. What is the correct classification of non-purchased?

0.90

3. What is the correct classification of purchased?

0.93

4. What is the correct and wrong classification of non-purchased?

0.96

5. What is the correct and wrong classification of purchased?

0.89

6. What is the overall performance of non-purchased?

0.93

7. What is the overall performance of purchased?

0.87

8. What is the average performance of correct and wrong classification?

0.89

9. What is the average performance of correct classification?

0.91

10. What is the average overall performance in classification?

0.90

11. What is the weight of correct and wrong classification?

0.91

12. What is the weight of correct classification?

0.91

13. What is the weight of overall performance in classification?

0.91

Support Vector Machine Classification Report

Class	Precision	Recall	F1-score	Support
0 (Non-Purchased)	0.77	0.97	0.86	79
1 (Purchased)	0.90	0.44	0.59	41
Accuracy				0.79
Macro Average	0.83	0.71	0.73	120
Weighted Average	0.81	0.79	0.77	120

1. What is the overall performance of the model Decision Tree?

0.79

2. What is the correct classification of non-purchased?

0.97

3. What is the correct classification of purchased?

0.44

4. What is the correct and wrong classification of non-purchased?

0.77

5. What is the correct and wrong classification of purchased?

0.90

6. What is the overall performance of non-purchased?

0.86

7. What is the overall performance of purchased?

0.59

8. What is the average performance of correct and wrong classification?

0.83

9. What is the average performance of correct classification?

0.71

10. What is the average overall performance in classification?

0.73

11. What is the weight of correct and wrong classification?

0.81

12. What is the weight of correct classification?

0.79

13. What is the weight of overall performance in classification?

0.77